Rushil Barve

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Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science (Co-op), Computational Mathematics — *Expected 2031*

Delhi Public School – Modern Indian School — Doha, Qatar
CBSE Senior Secondary, Computer Science — *Graduated 2026*

Projects

CircuitGuard | *Python, FastAPI, React, Three.js* — Sep 2026
- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.
- Python/FastAPI backend on the OpenF1 API, FAISS semantic search over FIA regulations, faster-whisper radio transcription, Next.js/React 3D track view.

Whisper Chain | *Godot 4, GDScript* — Oct 2026
- Puzzle game where each villager garbles a message (reverse, swap, double, drop a word); players route the message so the distortions cancel, discovering inverses and non-commutativity.
- Graph solver (DFS over simple paths) finds every correct route, sets par and powers live hints; used to design and validate all 10 levels.
- ~3,000 lines of GDScript with UI-free game logic, ~1,500 test assertions including randomized property tests, and GitHub Actions CI building web and Windows exports.

Daydream | *Godot 4, GDScript, Python, NumPy* — Oct 2026
- Puzzle game built on an inverted input mechanic: the dream world advances only while the player touches nothing; awake, they rearrange a desk whose side-on 3× projection is the dream level, so both worlds share one data model.
- Deterministic fixed-step (120 Hz) physics enables headless tests that replay a stored solution for all 8 levels and check each wins within par, run by a GitHub Actions workflow.

Lanternlight (in progress) | *TypeScript, React, Tailwind, Hono, Redis, Devvit* — Nov 2026 – Present
- Daily mirror-and-lantern puzzle for Reddit: a seeded generator builds each day's board, and an exhaustive solver over every reachable mirror arrangement sets the true minimum taps (par).
- Server (Hono + Redis) replays each player's tap sequence against the canonical puzzle to verify solves, then updates a sorted-set leaderboard, score distribution, streaks and stats; a scheduler posts a new puzzle daily.

MUN Chair Toolkit | *TypeScript, React, Vite, Tailwind, Zustand, Dexie* — Sep 2026
- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.
- Kept rules in framework-free functions (src/lib/rules.ts) with automated tests; deployed to GitHub Pages.

PayPilot (in progress) | *JavaScript, Node.js, Express, Claude API, PayPal REST APIs* — Nov 2026 – Present
- Chat agent that turns plain-English requests into PayPal invoices, payment links, reminders and refunds through a Claude tool-use loop over 11 PayPal API tools.
- Human-in-the-loop safety: actions that email customers or move money pause the agent loop, save its state and resume after an Approve/Decline click.

Door Occupancy Counter (in progress) | *Python, SQLite, JavaScript, TensorFlow.js* — Oct 2026 – Present
- Counts people entering and leaving each door from two-beam sensor events (direction from beam order), keeping a live headcount with no manual logging.
- Dashboard compares today with the same weekday over the last 4 weeks and forecasts the rest of the day; adds a busiest-hours heatmap, capacity alerts and an emergency roll-call view.
- Tracks its own accuracy from hand counts and empty-building nights; restores the day's count from the SQLite event log after a restart; daily Excel reports written with no external libraries.

Continental Smash | *Godot 4, GDScript, GLSL shaders* — Oct 2026
- Local two-player game where drifting continents ram each other to raise mountains and shove the rival into an ocean trench.
- Custom shaders for the planet, ocean, terrain and clouds, a narrated cinematic intro, particle impact effects and an Inspector-tunable ruleset.

Track Nector | *Python, HTML, JSON, Render* — Sep 2026
- Lightweight tracker for card balances and daily spending built around manual entry, with no bank account linking.
- Chose self-hosting and manual input deliberately to keep financial data private.

Corso Website | *HTML, Bootstrap, Sass, JavaScript* — Aug – Oct 2026
- Responsive site for courses and workshops: sticky navbar that changes on scroll, image carousel, registration and email-subscribe forms.
- Later cleaned up the course design myself: working forms with validation, accessibility fixes, ~70% smaller images; live at rsk0110.github.io/webproject.

Experience & Leadership

Waterloo Student Design Teams — Sep 2026 – Present
Member: WARG, UW Orbital, UW Game Dev — *Waterloo, ON*
- Waterloo Aerial Robotics Group (WARG): member of the Autonomy and Embedded Flight Software sub-teams.
- UW Orbital (student satellite design team) and UW Game Dev; also Computer Science Club and UW Data Science Club.

DPS-Modern Indian School May 2025 – Jun 2026
Director of MUN & Electoral and Democratic Club *Doha, Qatar*

- Led the school's Model UN program: organized conferences, trained delegates in research, procedure and resolution drafting.
- MUN trainer since May 2024; six Best Delegate awards across 8+ national and international conferences.

TEDxYouth@DPSMIS Sep – Dec 2025
Head of Marketing *Doha, Qatar*

- Led branding, social media and promotion for a TEDx-licensed youth event (11th edition, "Flip the Narrative").
- Worked across curation, design, operations and technical teams on the event campaign.

DPSDQ MUN (25th-anniversary edition) Jun – Oct 2025
Head of Operations *Doha, Qatar*

- Ran logistics and delegate services for the conference, coordinating with the Secretariat and support teams.

Qatar Children's Science Congress Dec 2024 – Mar 2025
Team Leader *Doha, Qatar*

- Led an original student research project from experimental design to data analysis and expert-panel presentation.

Catalyst MUN · Global MUN Network Jun 2025 – May 2026
Head Chair & Delegate Mentor *Doha, Qatar / Online*

- Head Chair of the Historical Security Council (Catalyst MUN) and the WHO (Global MUN Network); mentored first-time delegates.

HuAr (Humanities.Arts) Feb 2024 – Mar 2025
Co-Founder *Doha, Qatar*

- Co-founded a student organization guiding students on humanities and arts paths; organized conferences and speaker sessions.

Q-SOMA (Qatar Society of Medical Aspirants) Jun 2025 – Jun 2026
Head of Outreach & Logistics *Doha, Qatar*

- Led outreach and logistics and supported event management for the society.

New Gen Network Aug – Nov 2025
Event Manager *Doha, Qatar*

SFI Football Cup (StraightForwardInsight) May 2025
Event Coordinator *Doha, Qatar*

- Coordinated logistics and operations for a charity football cup.

AetherMUN · Luminos Foundation May – Aug 2025
Content Contributor · Brand Ambassador *Online*

DPS-Modern Indian School May 2024 – Jan 2025
Education Mentor *Doha, Qatar*

- Mentored a junior student while in Grade 11.

CMU-Qatar MindCraft Workshop · Embassy of India Jun – Nov 2024
Student Volunteer *Doha, Qatar*

- Assisted students with robotics activities; volunteered at the International Day of Yoga.

Hackathons & Awards

- 8 hackathons, 5 won, 1 organised: FormulaTech Hacks 2026 (Waterloo; Winner, Ampere: AI for Motor Safety), RevenueCat Shipaton, OpenCV AI Competition (AWS), Nebius x NVIDIA Global AI Hackathon, IBM BOB 2.0 (all 2026), DPS MIS Doha (2025), Birla School (2024), CMU Qatar (2023).
- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze Medal of Distinction (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Special Achievement Certificate, International Olympiad of English Language (2024-25)
- Best Delegate, UN Commission on Crime Prevention and Criminal Justice, Monarch MUN (2024-25)
- Honorary Mention, SDG 13, DPS-Modern Indian School MUN (2024-25)
- Six Best Delegate awards and one Honorable Mention across 8+ conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26); Bronze (2023-24, 2024-25); national-level athlete

Technical Skills

Languages: Python, JavaScript, TypeScript, GDScript, C, SQL, HTML, CSS
Frameworks: React, Vite, Tailwind CSS, FastAPI, Node.js, Express, Hono, Godot 4, Zustand
Tools: Git, GitHub Actions, SQLite, IndexedDB, Render, Redis, Reddit Devvit, Claude API, PayPal REST APIs
Certifications: Python Essentials 2 · HTML Essentials · Computer Hardware Basics; Accenture Nordics – Software Engineering Job Simulation; Bloomberg – Client Service Job Simulation